DEFENSE RESEARCH

Defensive Run Value Level Translations

What minor league defensive numbers tell us about the next level.

By Inaki Sardina

October 10, 2026

SFG defensive run value · 2021–2026

Games through August 21, 2026

Project Summary

A player's defensive numbers shrink with every promotion, but where they rank among their peers holds up. Good defenders in the minors usually become good defenders in the majors.

0.30

runs: a one-run advantage in FRV/100 at one level corresponds to approximately a 0.30-run advantage at the next level, on average (0.32 for catcher framing).

67%

of top-20% position players stay above average after moving up (71% of top-20% framers).

72%

of today's above-average MLB position players were above average in the minors (85% of framers).

16%

of the difference in catchers' season value comes from throwing, with framing the rest.

Introduction

When a player moves up a level, how much of their defensive value comes with them? This report answers that using SFG defensive run value from rookie ball through MLB.

Project Questions:

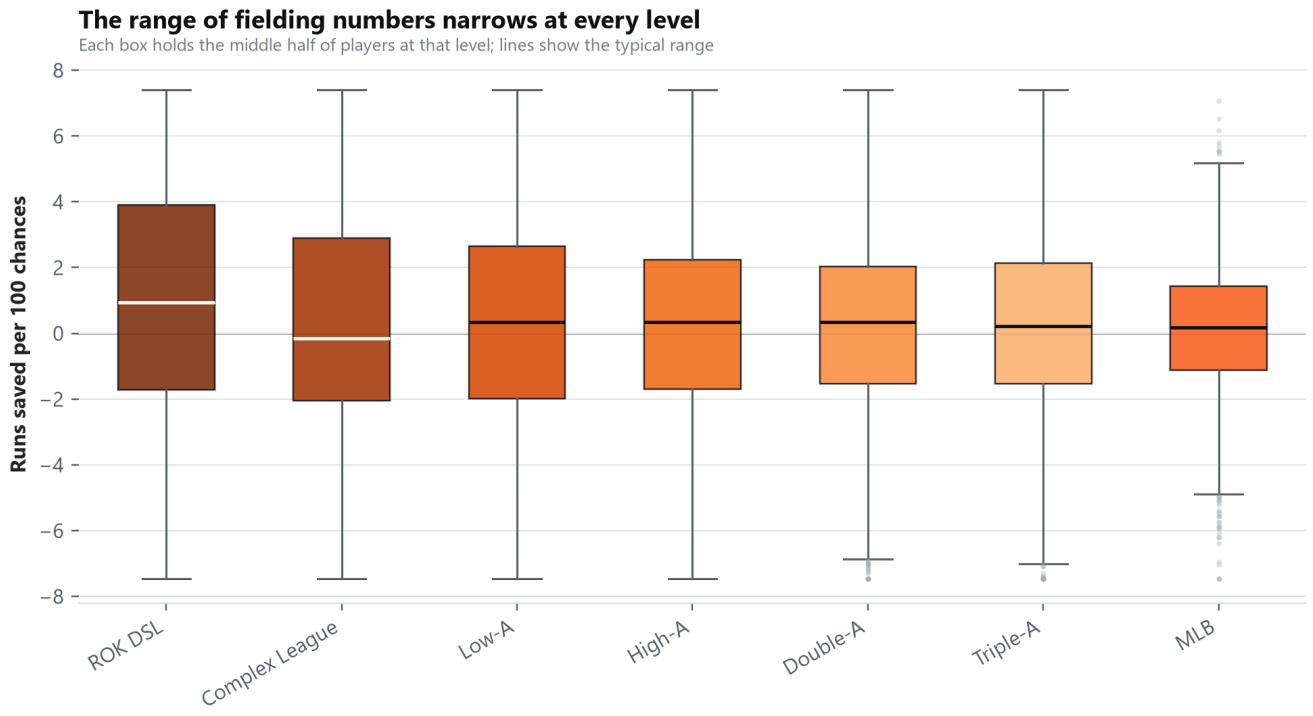
1. How do defensive numbers change as players move up the ladder?
2. How much of a player's defense at one level shows up at the next?
3. How do catching skills translate across levels?

How to read the numbers:

- **Runs saved per 100 chances** measures a player's defense. 0 is average, positive is better than average, negative is worse. For position players a chance is a ball in play they could make a play on; for catchers chances are called pitches and basestealing throwing opportunities.
- **Rank among peers** compares a player only with others at the same level, in the same season, at the same position. A player at 80% ranks ahead of 80% of their peers.

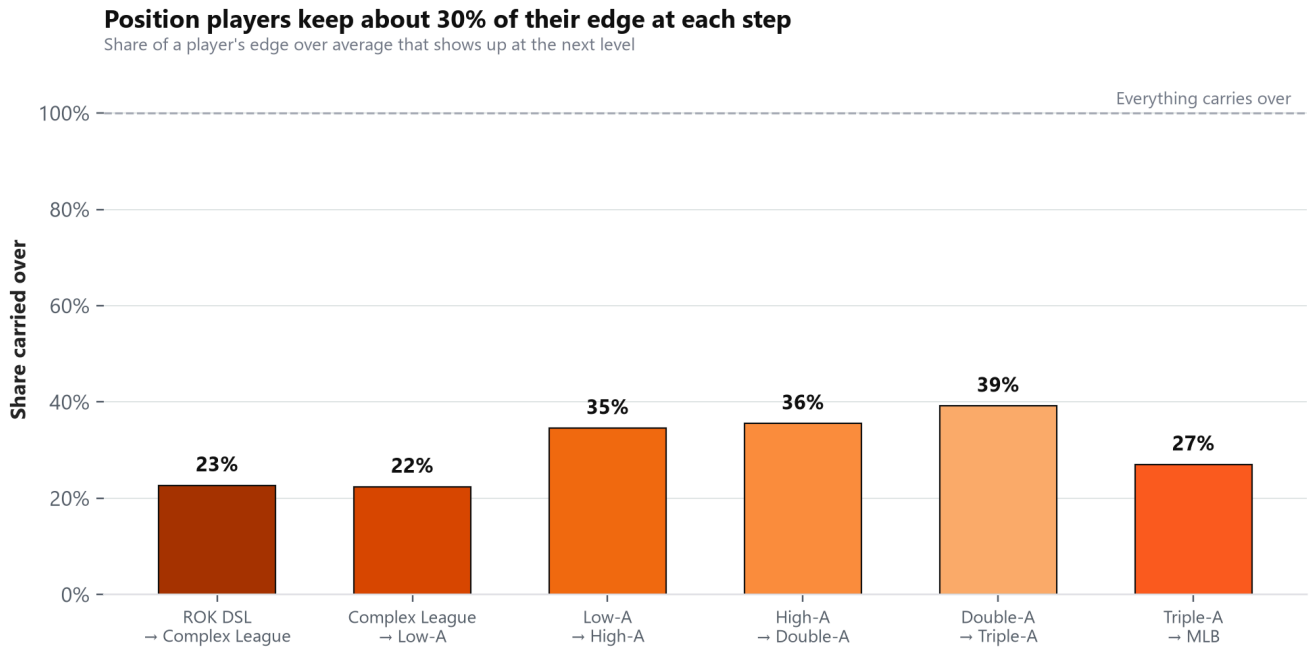
Position Players

Rookie-ball numbers swing much more



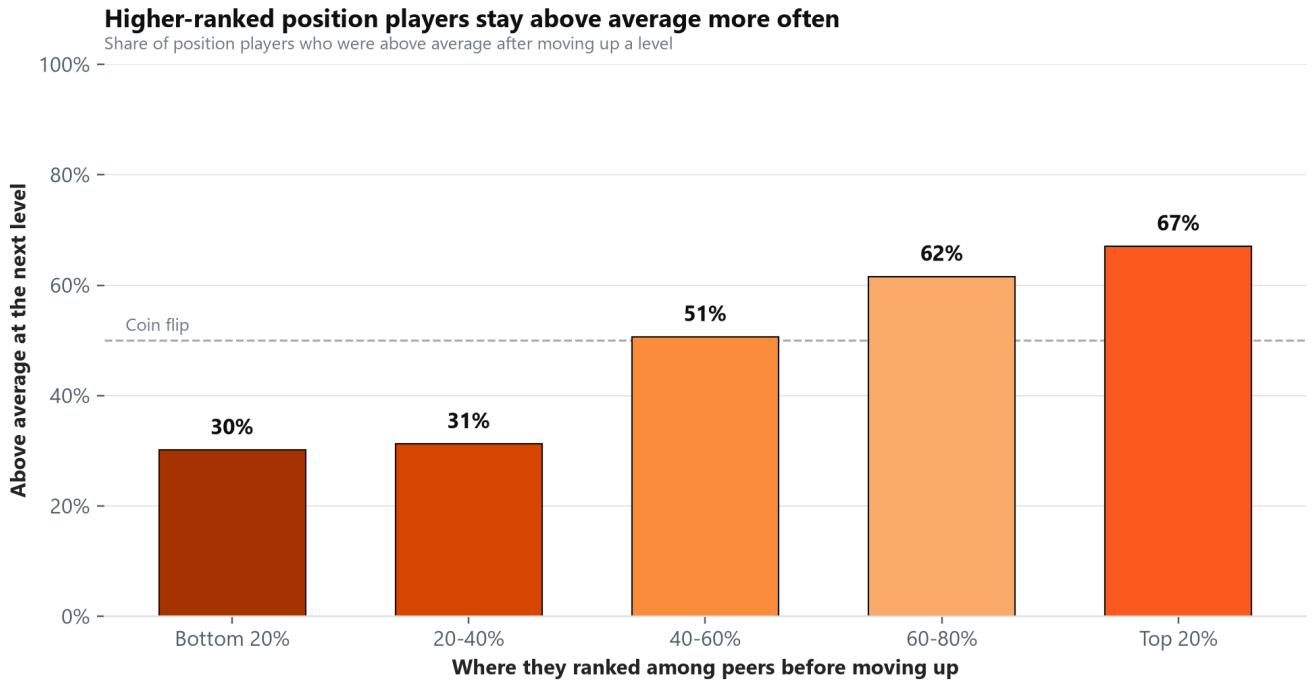
Rookie-ball fielding numbers are spread much wider than MLB numbers. The range covering most players (the middle 80%) is 2.3 times as wide in ROK DSL as it is in MLB. Players get fewer chances in rookie ball, talent is more uneven and tracking systems are less reliable, so big numbers in either direction are more common and “mean” less.

Only part of a player’s edge moves up with them



When a position player moves up a level, only about 30% of their edge over an average player shows up at the next level. For every 10 runs above average at one level, expect roughly 3 above average at the next. The pattern holds at every step, from Rookie Ball to Triple-A to MLB.

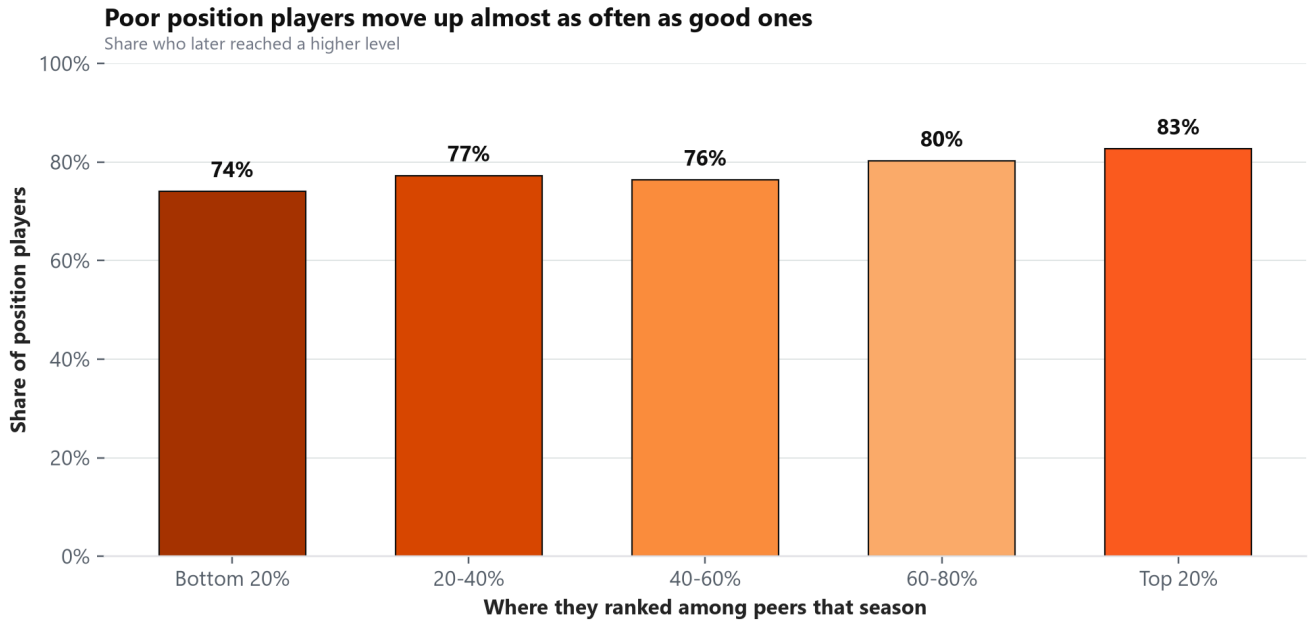
Where a player ranks carries forward

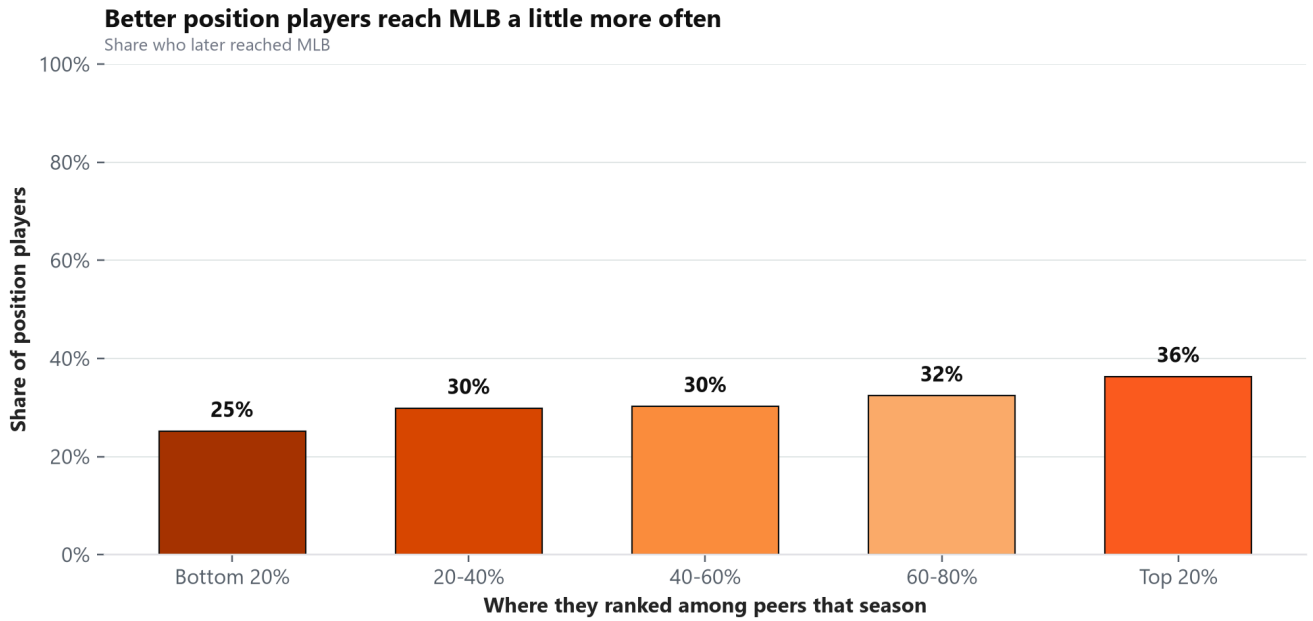


How to read this chart: position players are sorted into five groups by where they ranked among their peers before a promotion, from the bottom 20% on the left to the top 20% on the right. Each bar shows how many in that group were **above average at the next level**. If rank meant nothing, every bar would sit near the 50% line, like a coin flip.

Instead, the bars climb from left to right. Position players in the top 20% are above average after moving up 67% of the time, while position players in the bottom 20% get there only 30% of the time. Even though the raw numbers shrink, a player's rank holds up.

Do poor position players still move up?

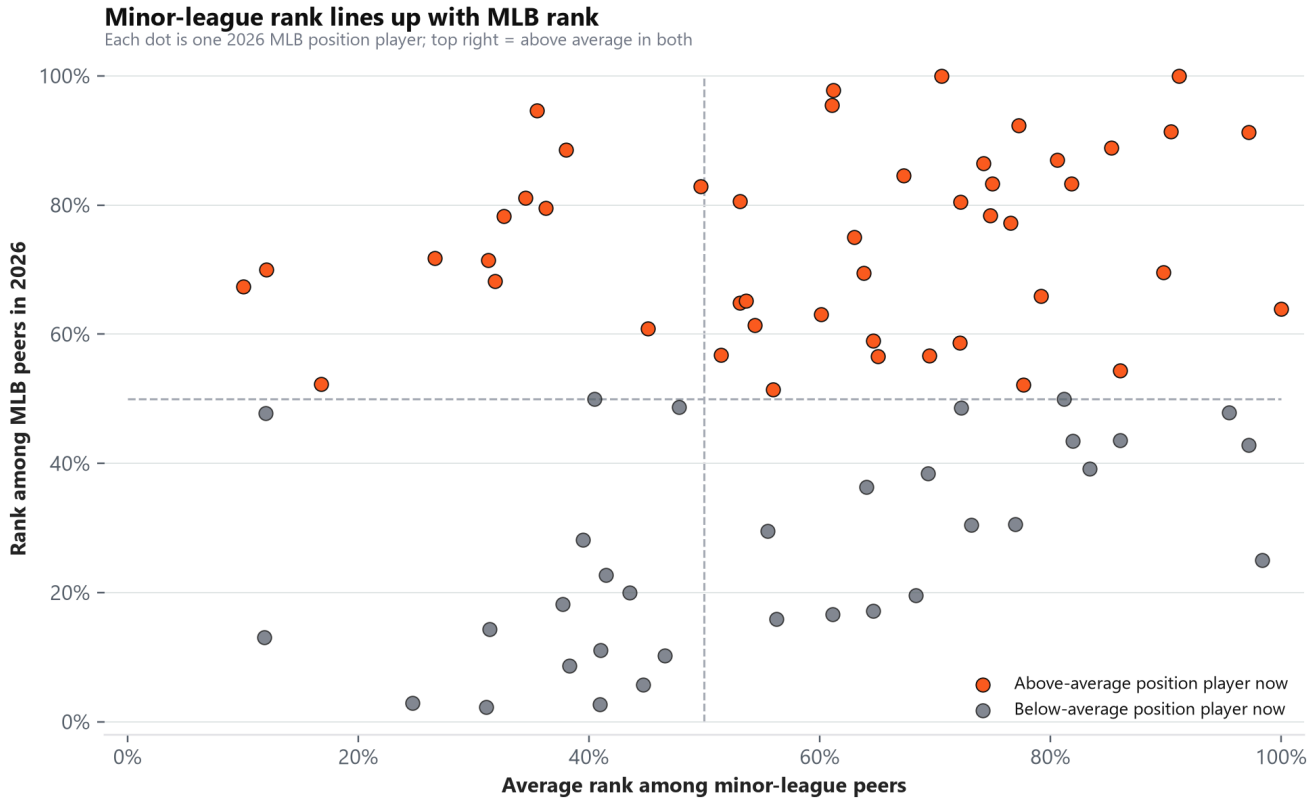




Yes, most of them do. Looking at minor-league seasons through 2024 (so every player has had at least two more seasons to move up), position players who ranked in the bottom 20% at their level still moved up at least one level 74% of the time, compared with 83% for the top 20%. 25% of bottom-20% position players eventually reached MLB, compared with 36% of the top 20%.

Defensive performance is associated with promotion rates, but it is not the sole determining factor. Players with below-average defense frequently advance to higher levels, where their defensive struggles can persist.

Today’s good MLB position players were good in the minors

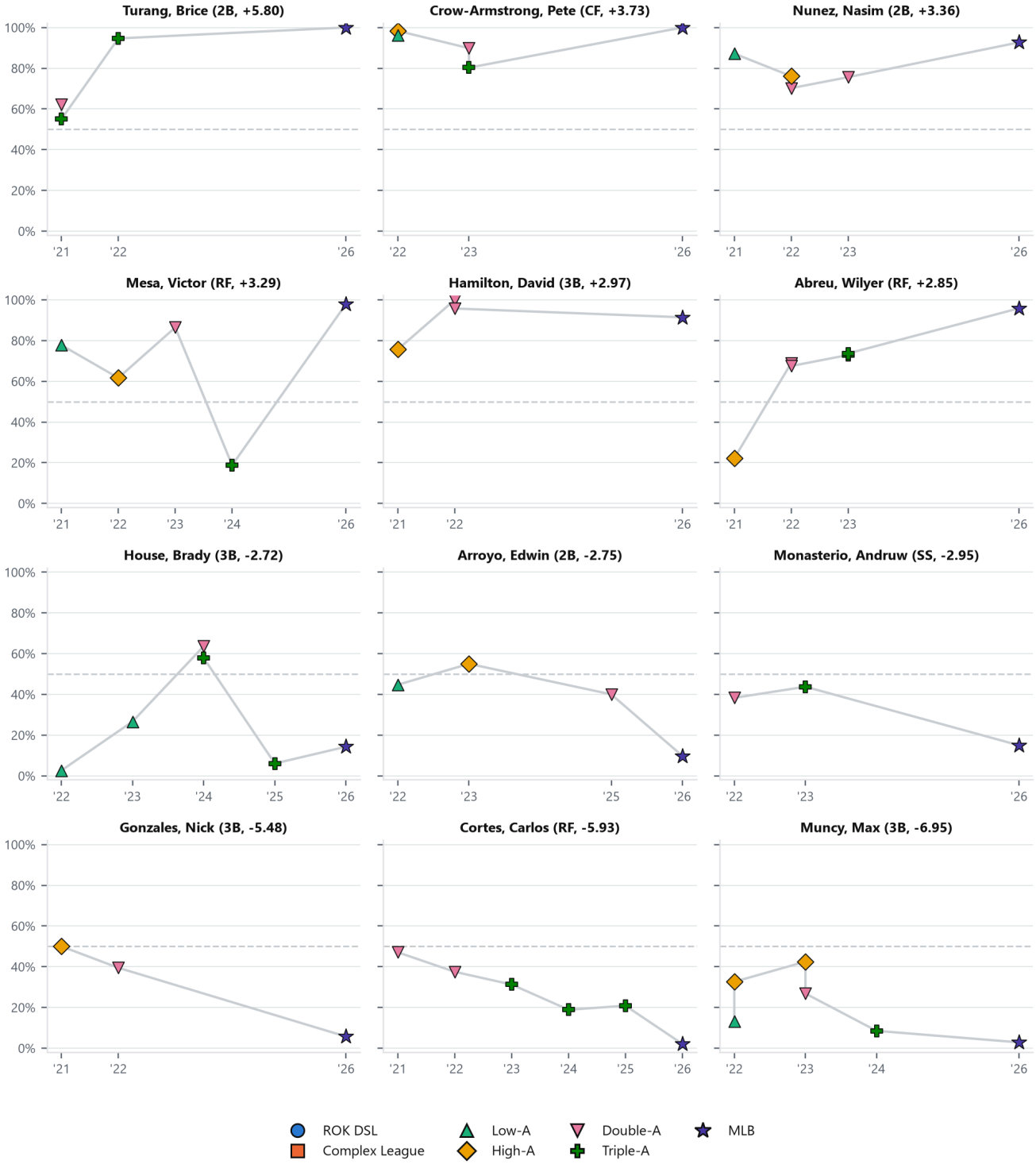


Looking back at 2026 MLB position players who came up through the minors recently, 72% of the above-average ones were also above average in the minors. A good minor-league track record doesn’t guarantee a good MLB defender, but it improves the odds.

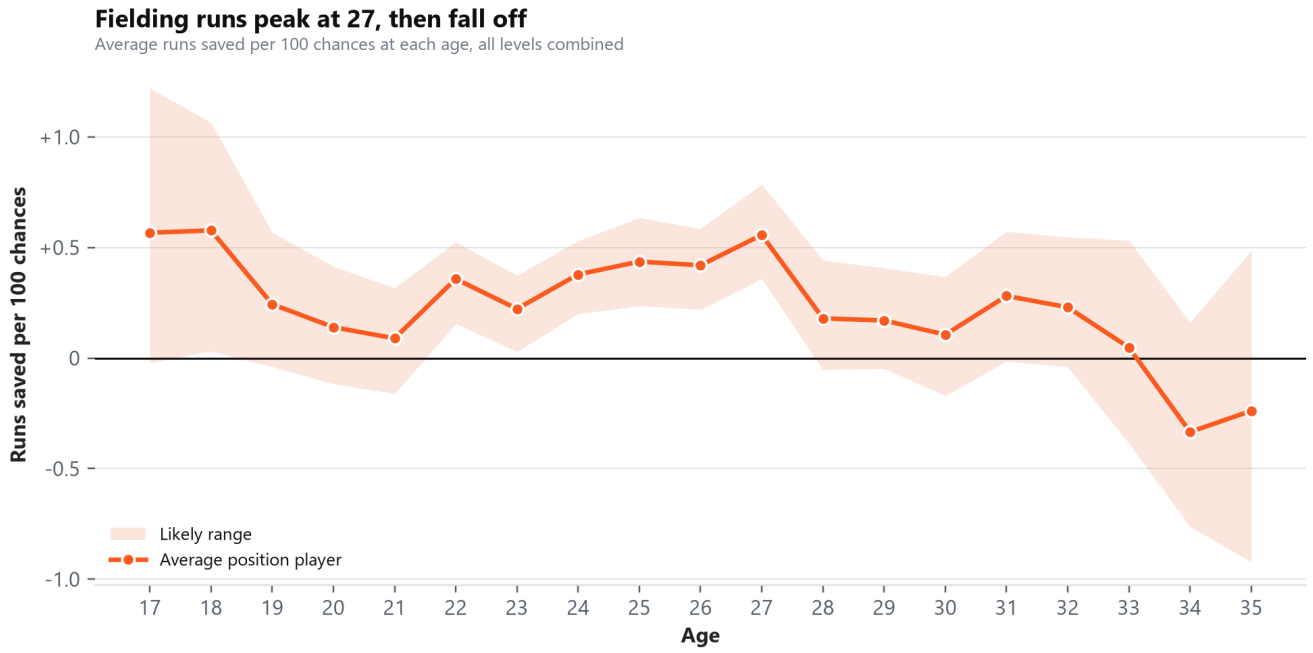
The six best and six worst of those MLB position players this season, and how they ranked at each stop on the way up:

Position player paths to MLB: the best and worst of 2026

Rank among peers at each stop on the way to MLB (100% = best at that level).
Number next to each name = runs saved per 100 chances in 2026.



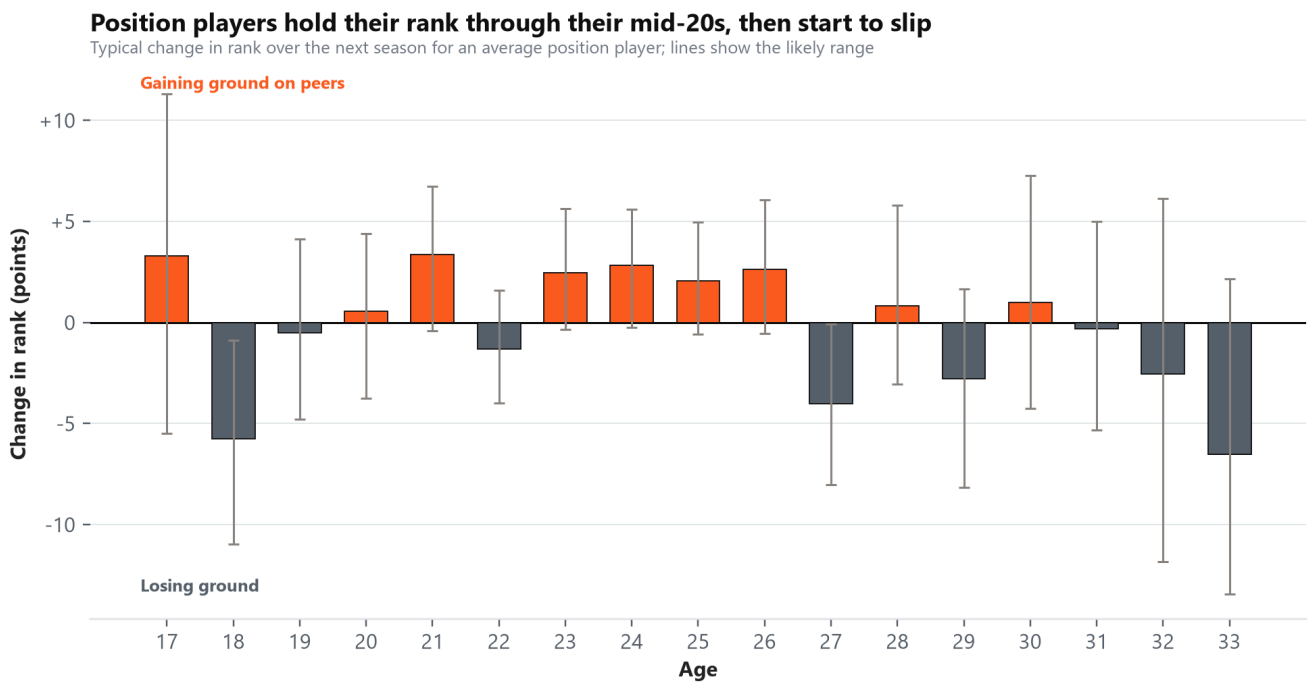
Age: position players keep pace until their late 20s



Runs saved at each age. First, the plain numbers: the average runs saved per 100 chances for position players at each age, across all levels.

- **Building up:** after a low point at 21 (+0.09 runs per 100 chances), fielding runs climb to a peak of +0.56 at age 27.
- **Falling off:** from 28 on, the average drops to +0.14, and by 34 the typical position player is below zero.
- **The youngest ages** (17–18) look strong too, but most of those seasons are in rookie ball, where numbers swing the most.

Players of the same age are spread across different levels, so this chart mixes getting older with who is playing where. The next chart handles that by comparing each player only with peers at their own level.

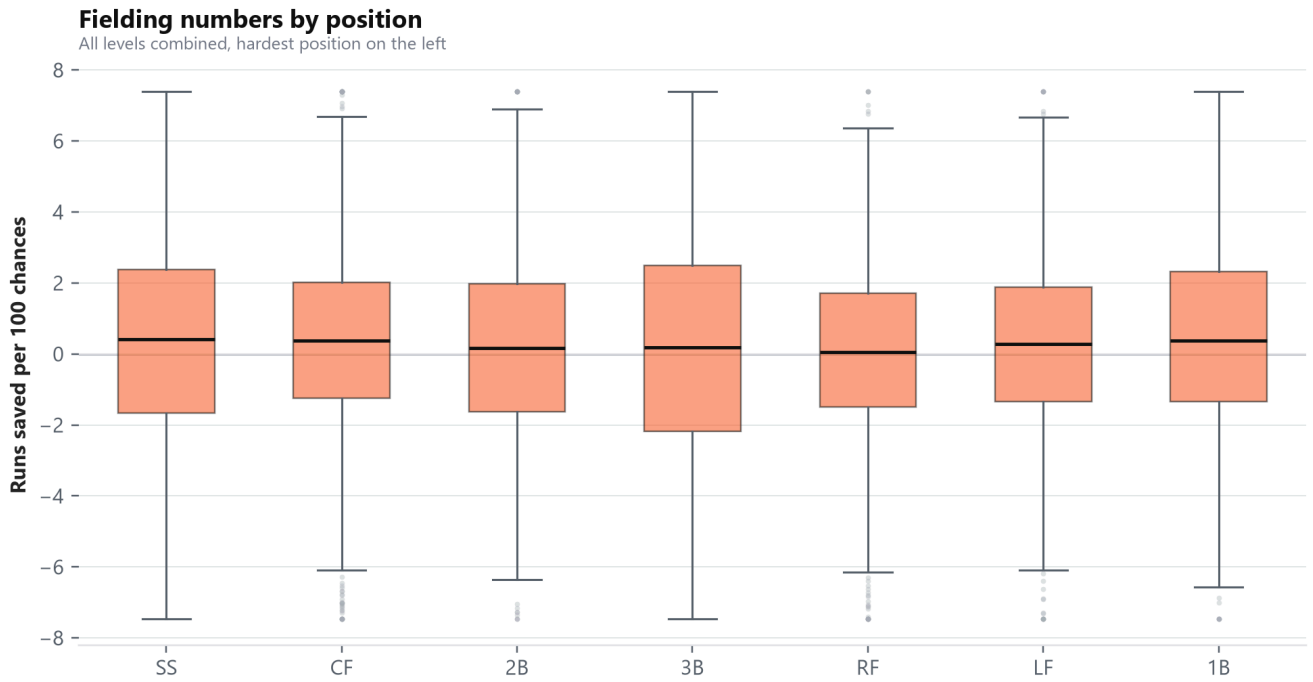


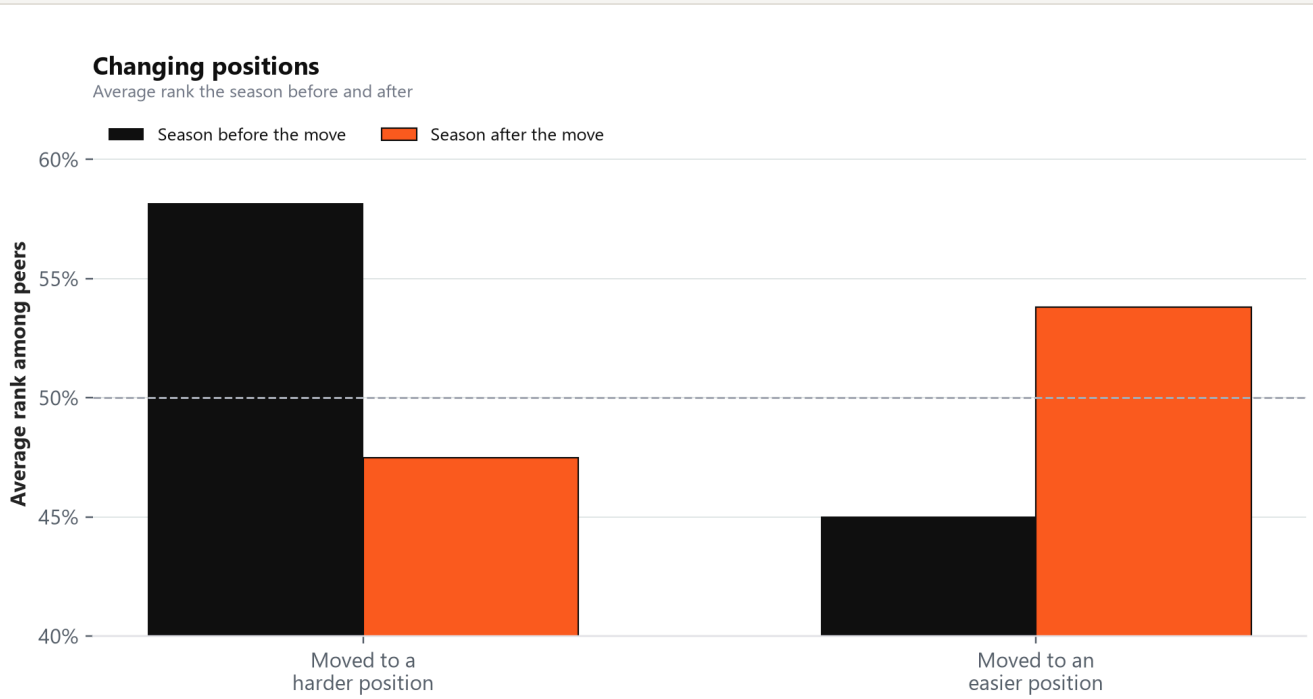
Keeping pace with peers. This chart asks how a position player's **rank among their peers** changes from one season to the next as they get older. Because rank is measured against players at the same level, players can be followed even when they move up. Each bar is the typical change for an average position player at that age. Above zero means they gained ground on their peers over the next season; below zero means they lost ground.

- **Up to 26:** position players hold or slightly gain ground, about +1.3 rank points a season on average.
- **From 27:** they tend to slip, about -1.8 points a season, with the biggest drops in the early 30s.

The lines on each bar show how uncertain that single age is, and most of them cross zero. The pattern across ages is more reliable than any one bar.

Position: players grade worse at harder spots





Same player, two positions. When a player played two positions in the same season and at the same level, they graded worse at the harder position 63% of the time. The biggest gap was

between CF and RF.

Changing positions. Players who moved to an easier position ranked about 9 points higher the next season. Players who moved to a harder position ranked about 11 points lower.

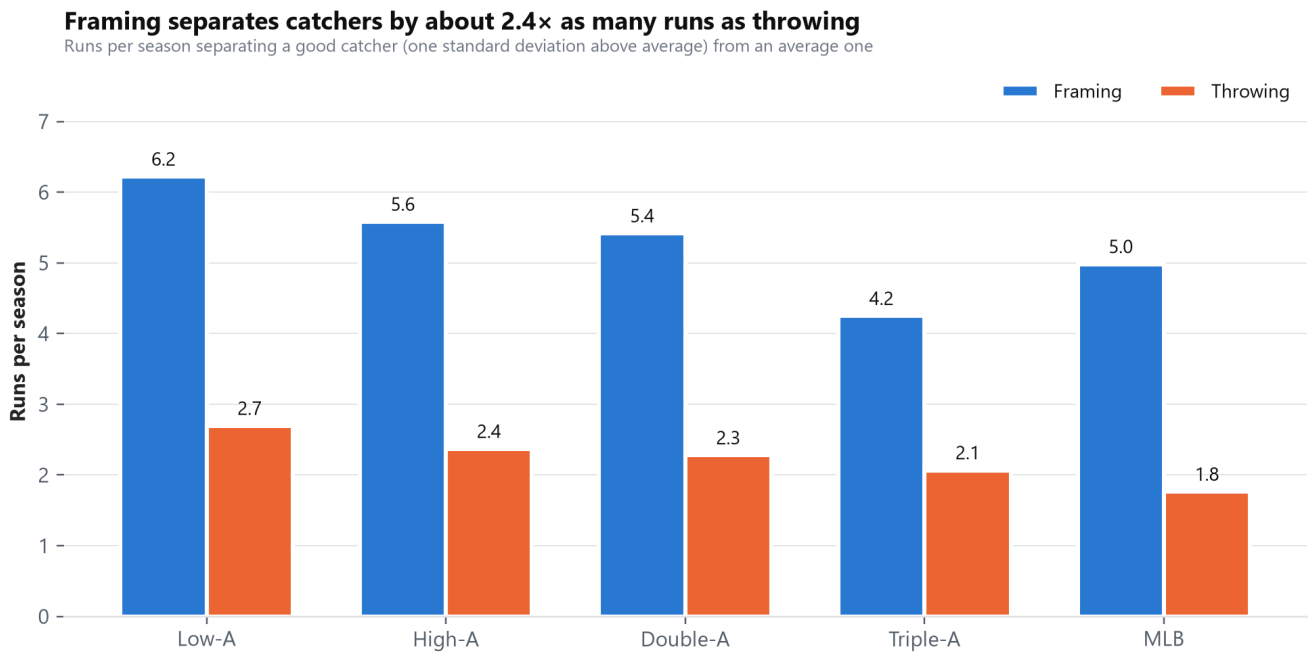
Position player takeaways

- Rookie-ball fielding numbers swing widely, so judge players against their peers at that level.
- Expect about 30% of a position player's edge to carry to the next level.
- Rank carries forward: top-20% position players usually stay above average after moving up.
- Poor position players still get promoted most of the time, so their defensive problems move up with them.
- Position players keep pace with their peers through their mid-20s and start to slip from their late 20s.
- Position matters: expect a player's grade to drop at a harder position and rise at an easier one.

Catchers

Catchers generate most of their defensive value differently from other position players, primarily through **framing** and **throwing**. Throwing data covers 2016–2026 from Low-A to MLB.

Framing matters most, but throwing counts too

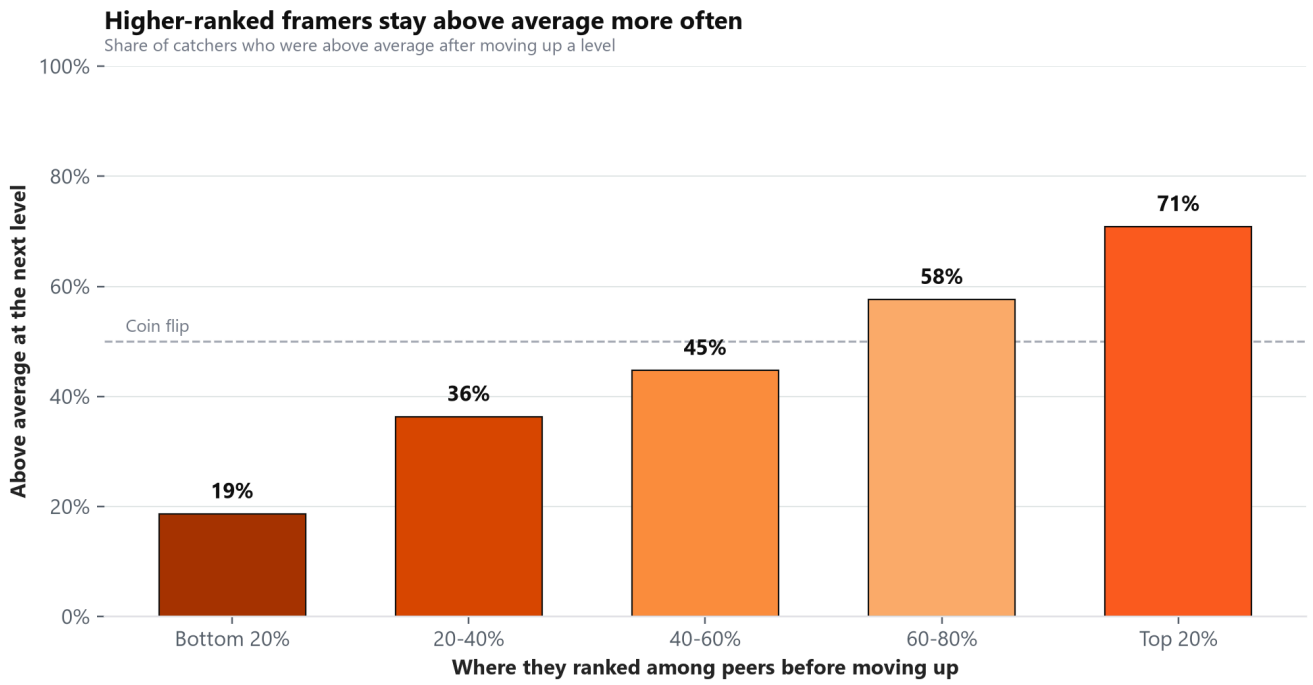


Framing happens on every pitch and throwing only on steal attempts, and over a season framing makes the bigger difference. A good framer (one standard deviation above average) saves about 5.0 runs a season in MLB, and a good thrower about 1.8. Across levels, throwing accounts for 11%–19% of the spread in catchers' combined value, with framing the rest.

- **They're separate skills.** A catcher's framing and throwing runs are barely related (correlations of 0.02 to 0.15 by level), so a good framer isn't necessarily a good thrower. Each needs its own look.

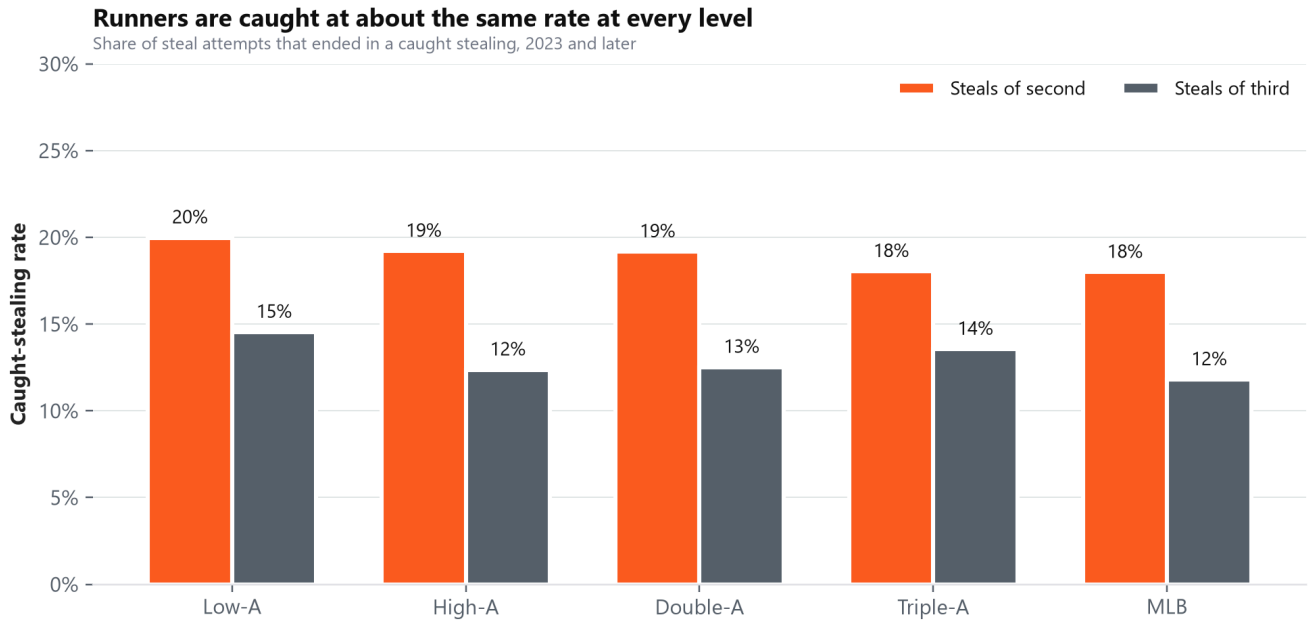
Framing: the rank travels even when the number doesn't

As with position players, framing numbers swing much more in rookie ball (the spread is 3.8 times as wide in ROK DSL as in MLB), and a catcher keeps only about 32% of their framing edge when they move up a level. The Run Value Translator below covers the numbers for any pair of levels. What stands out for catchers is how well **framing rank** holds up:



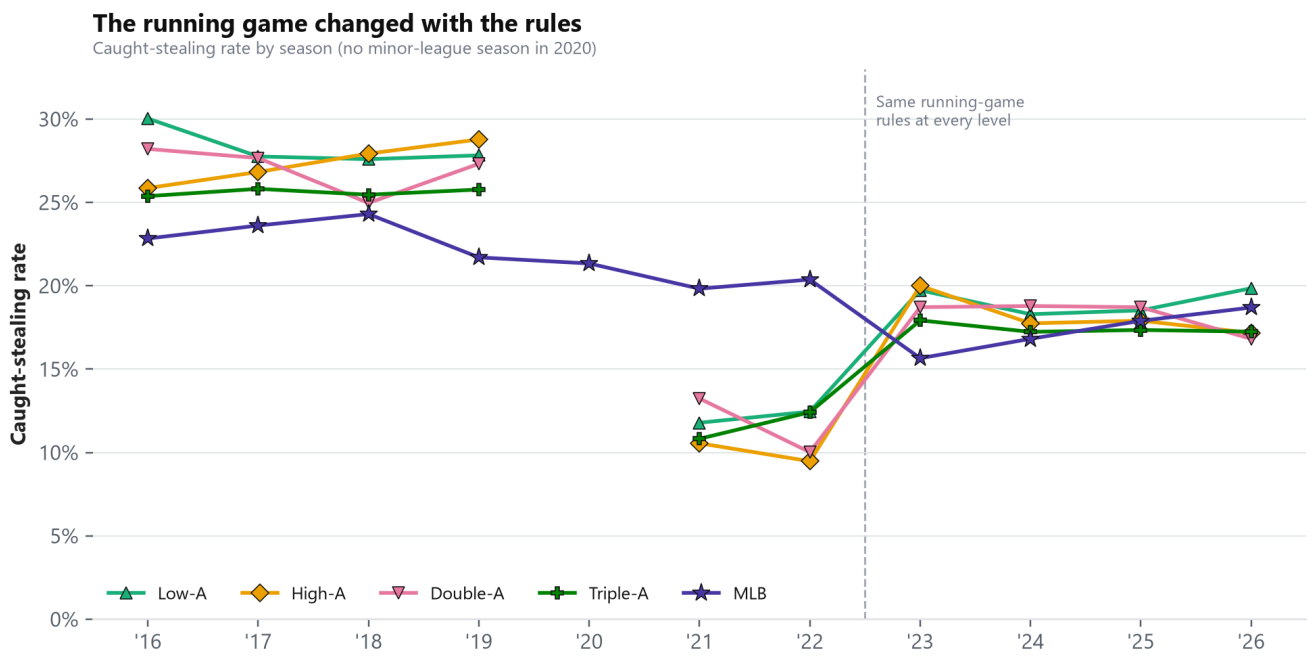
Catchers in the top 20% at their level are above average framers at the next level 71% of the time, and those in the bottom 20% only 19% of the time. That's a wider gap than for position players (67% against 30%), making framing rank one of the steadiest signals in this report. It shows at the top too: 85% of today's above-average MLB framers were above-average framers in the minors.

Caught-stealing rates across levels



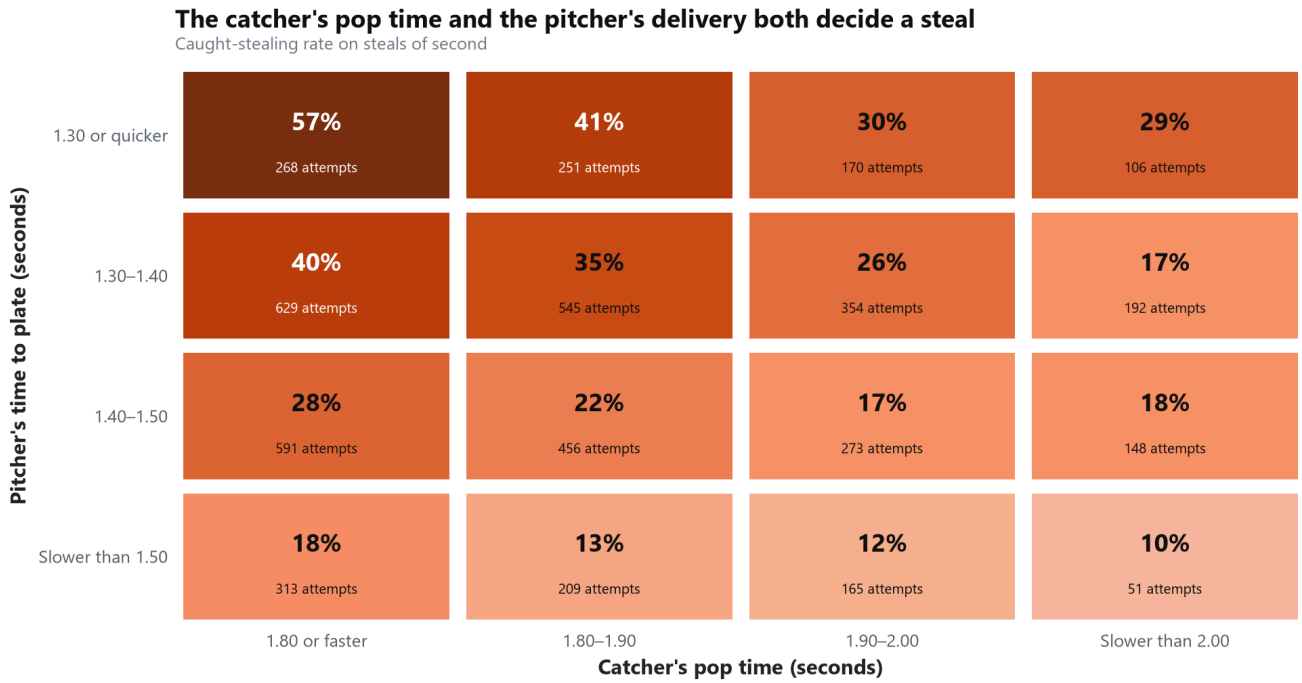
Since 2023, when MLB adopted the running-game rules already in use in the minors (bigger bases and limits on pickoffs), runners have been caught about as often at every level: 17%–19% of attempts.

- **Steals of second are caught far more often than steals of third:** 18%–20% against 12%–15%. Steals of third are rarer (15% of attempts), likely because runners are more cautious when attempting a steal of third.
- **The level doesn't change the odds much,** even though pop times get quicker higher up (see below), suggesting runners may be more selective, attempting them only when success is more likely.



Over time, caught-stealing rates fell. Before 2020, minor-league runners were caught 27% of the time and MLB runners 23%; since 2023 it's 18% in the minors and 17% in MLB. The 2021–2022 minor-league seasons, when the new rules were being tested, show unusually low rates (11%); treat those two seasons with care, since they may partly reflect how those games were recorded. Comparisons between levels in this section use 2023 and later.

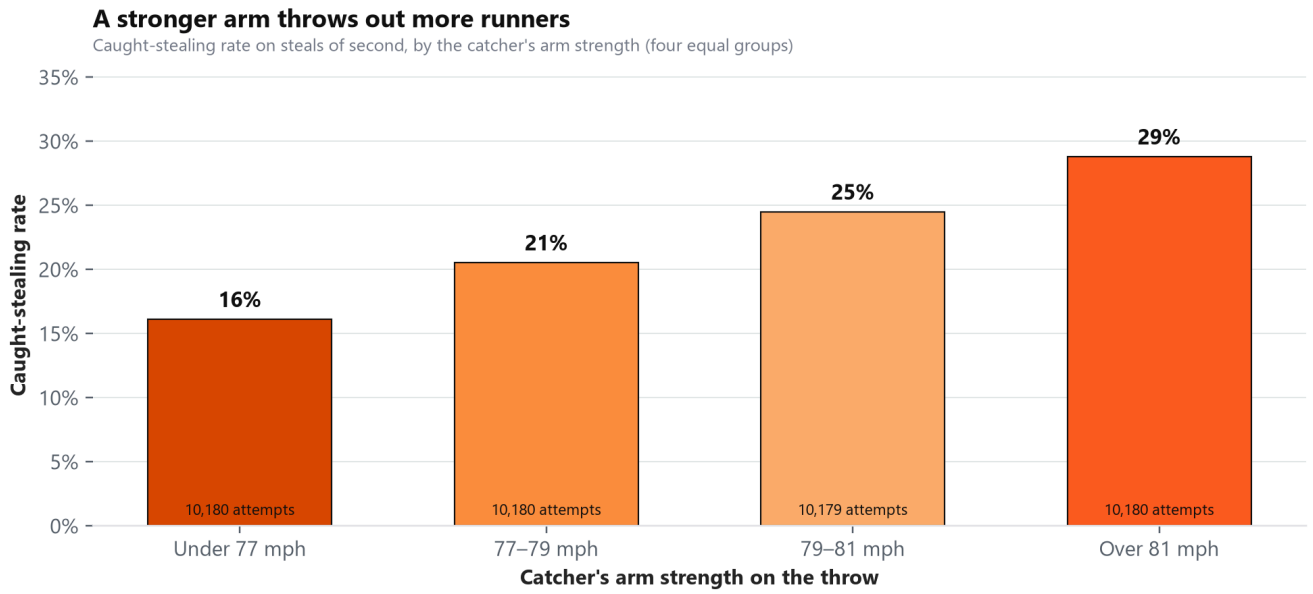
What decides a Successful Steal



Each cell shows how often runners were caught stealing second, by the catcher's pop time (across) and the pitcher's time to the plate (down). A fast pop behind a quick pitcher throws out 57% of runners; a slow pop behind a slow pitcher, 10%.

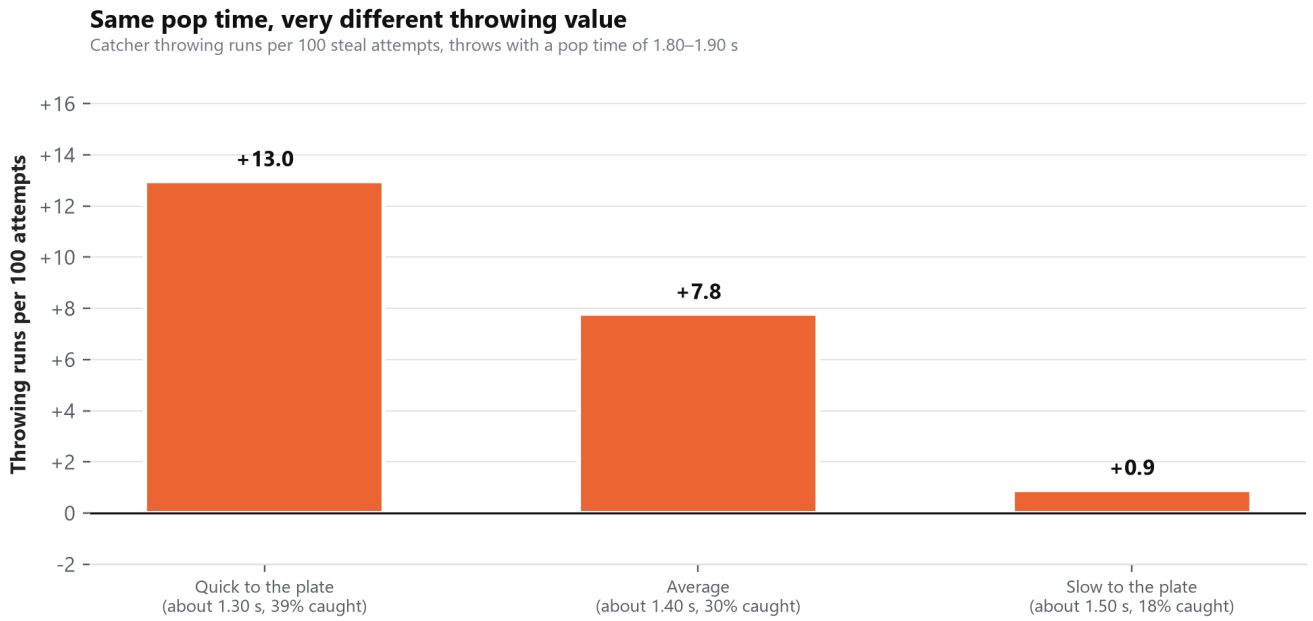
Looking at the catcher's pop time and arm strength, the pitcher's delivery time and the runner's speed together on the same attempts, a typical difference in each moves the caught-stealing rate by:

- **Pitcher's time to the plate:** about 11 points for 0.10 s quicker, the biggest of the four.
- **Runner speed:** about 9 points for a runner 0.9 ft/s faster (in the runner's favor).
- **Catcher's pop time:** about 5 points for 0.13 s faster, or 4 points per tenth of a second.
- **Catcher's arm strength:** about 5 points for a throw 3.3 mph harder, or about 1.4 points per mph.



Arm strength and pop time are the two parts the catcher controls, and they're related but not the same thing (correlation -0.35). Pop time also includes the transfer and footwork, so a catcher with a lighter arm can make up some ground with a quick release, and a strong arm can be wasted by a slow one. On the same throws, each is worth about the same. Catchers with the weakest arms throw out 16% of runners, against 29% for the strongest. Even so, the pitcher's delivery matters at least as much as either.

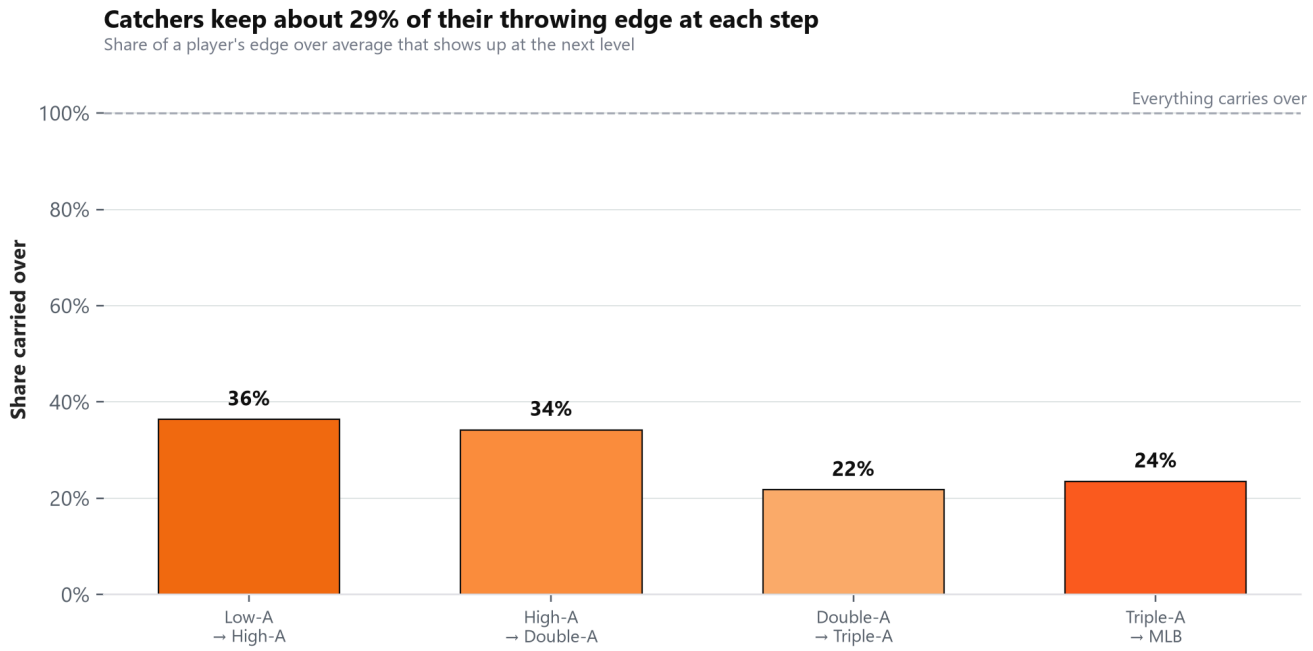
Part of a catcher’s throwing value belongs to the pitcher



To isolate the pitcher’s part, this compares throws with nearly the same pop time (1.80–1.90 s). Behind quick-to-the-plate pitchers, catchers earn +13.0 throwing runs per 100 attempts; behind slow ones, +0.9. The catcher’s pop time was about the same; the pitcher’s delivery made most of the difference.

So a catcher’s throwing runs partly reflect the staff they catch. When comparing catchers’ throwing, check the time to the plate of the pitchers they worked with, and lean on pop time for the catcher’s own part.

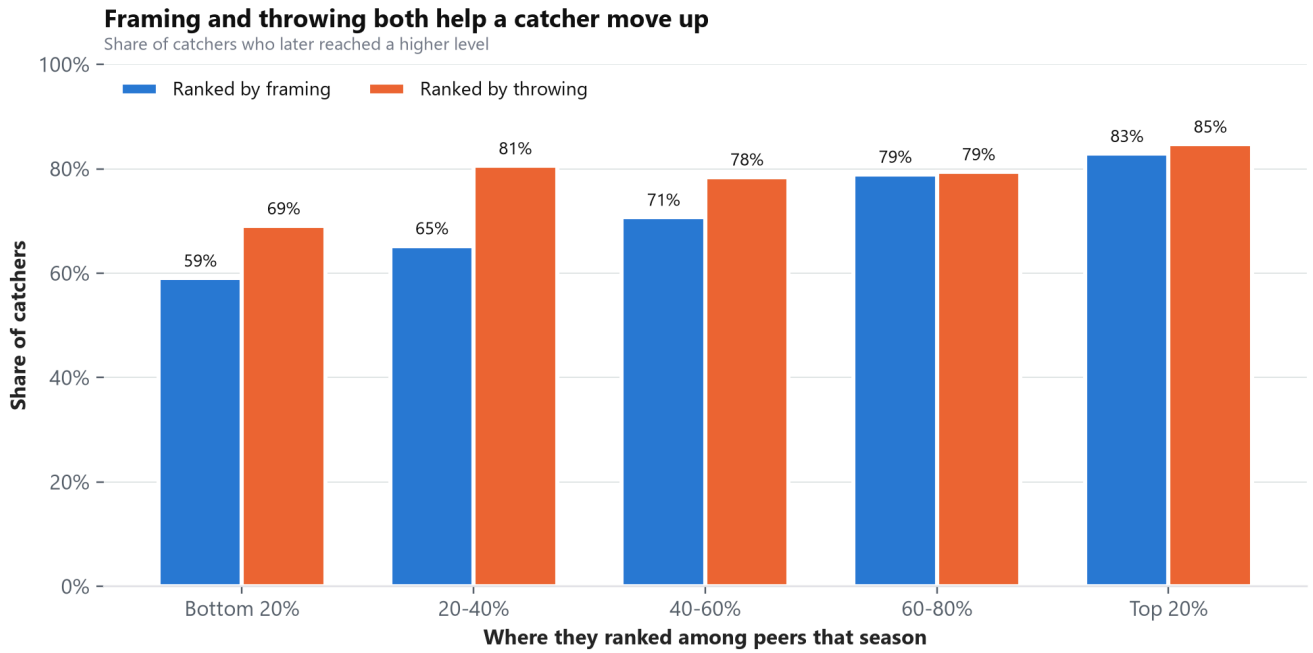
Throwing up the ladder



A catcher keeps about 29% of their throwing edge at each step up, about the same as the 32% they keep for framing. Throwing rank holds up less well than framing rank: top-20% throwers are above average at the next level 73% of the time, against 37% for the bottom 20% (for framing, 71% against 19%). A catcher also gets a new pitching staff at every level, which moves their throwing numbers.

- **Pop time gets quicker up the ladder.** The typical catcher's pop time goes from 1.96 s in Low-A to 1.87 s in MLB, while runners are caught about 20% of the time at every level.
- **Pop time travels better than throwing value.** For the 216 catchers with tracked throws at two neighbouring levels, their pop time at the higher level kept about 67% of its difference from average, against 29% for throwing value. That makes pop time a steadier read on a catcher's arm than the throwing value.

Who moves up



Catchers who rank well at either skill are more likely to move up. By framing, the top 20% moved up 83% of the time against 59% for the bottom 20%. By throwing, it was 85% against 69%. Framing also shows in who reaches MLB: 24% of top-20% framers got there, against 10% of the bottom 20%.

Catcher takeaways

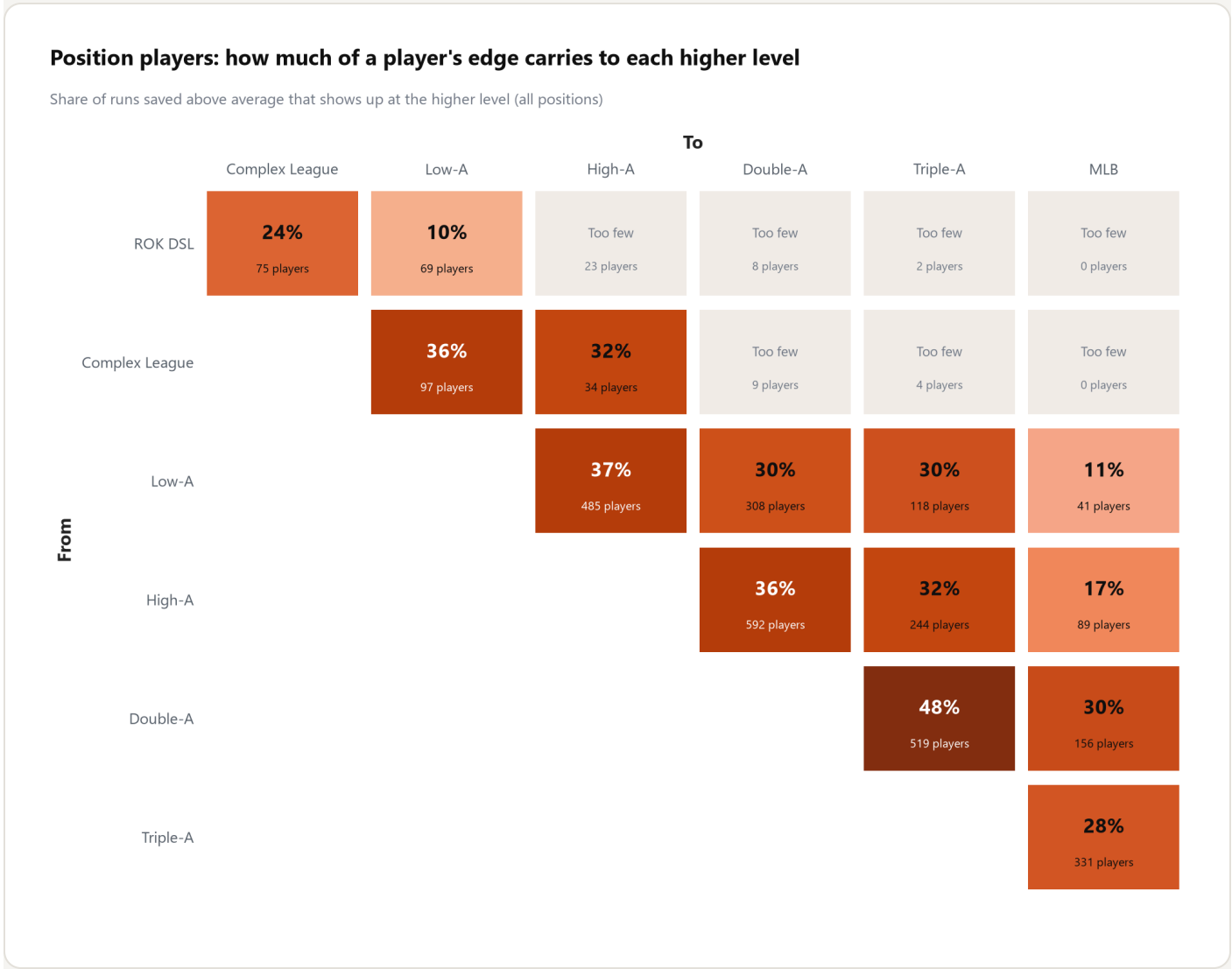
- Framing separates catchers by about 2.4× as many runs as throwing over a season, but the two are separate skills. Look at both.
- Framing rank is one of the steadiest signals in this report: good framers usually stay good as they climb.
- The pitcher's time to the plate moves steal outcomes at least as much as pop time does, and it shows up in a catcher's throwing runs. Judge a catcher's arm by pop time and in the context of their staff.
- Pop time is a steadier read on a catcher's arm than the throwing value: it gets quicker up the ladder and carries better from level to level.

Run Value Translator

If a player saves X runs per 100 chances at one level, what should we expect at a higher level? And what if a position player moves to the position next door? This section answers both, using players who actually made the move.

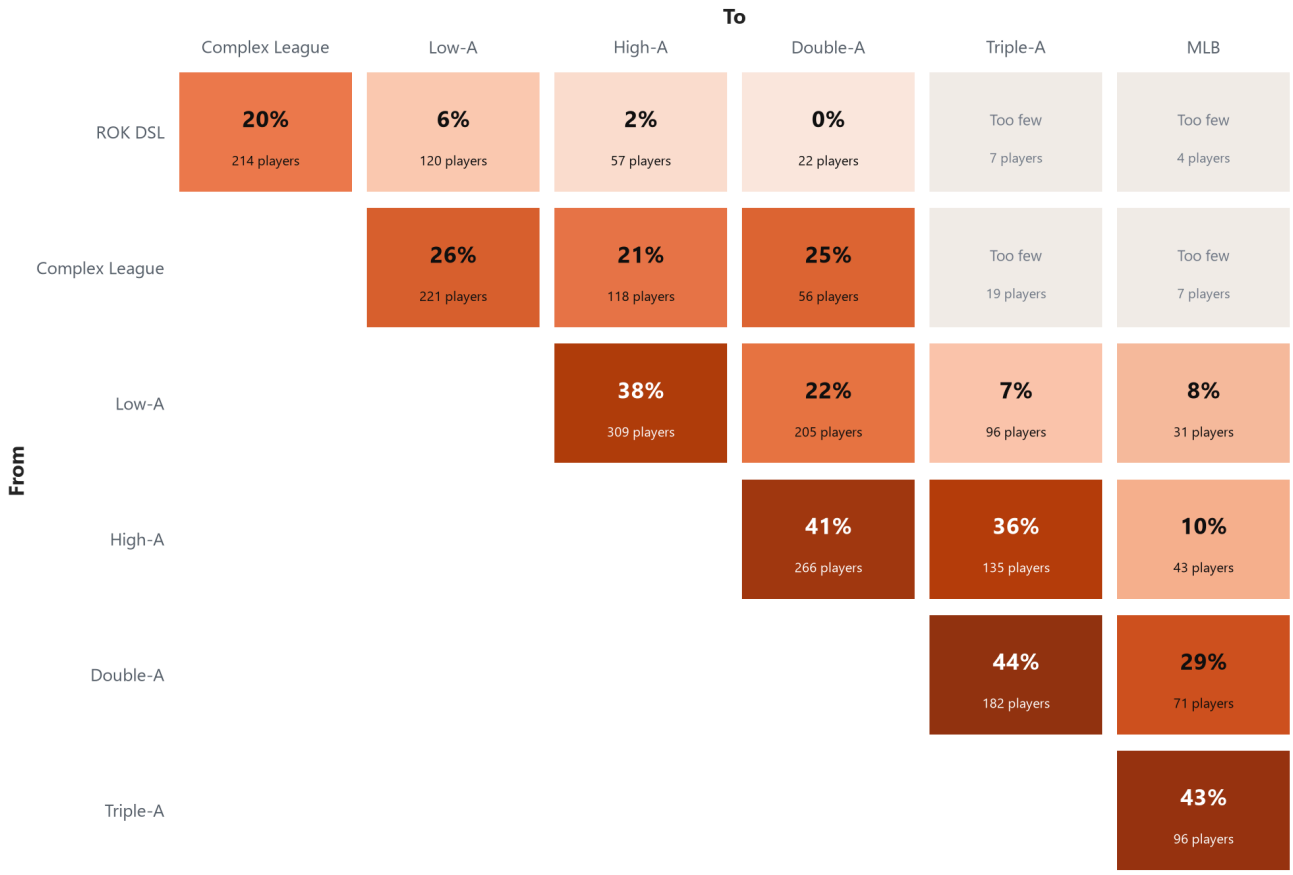
Moving up a level

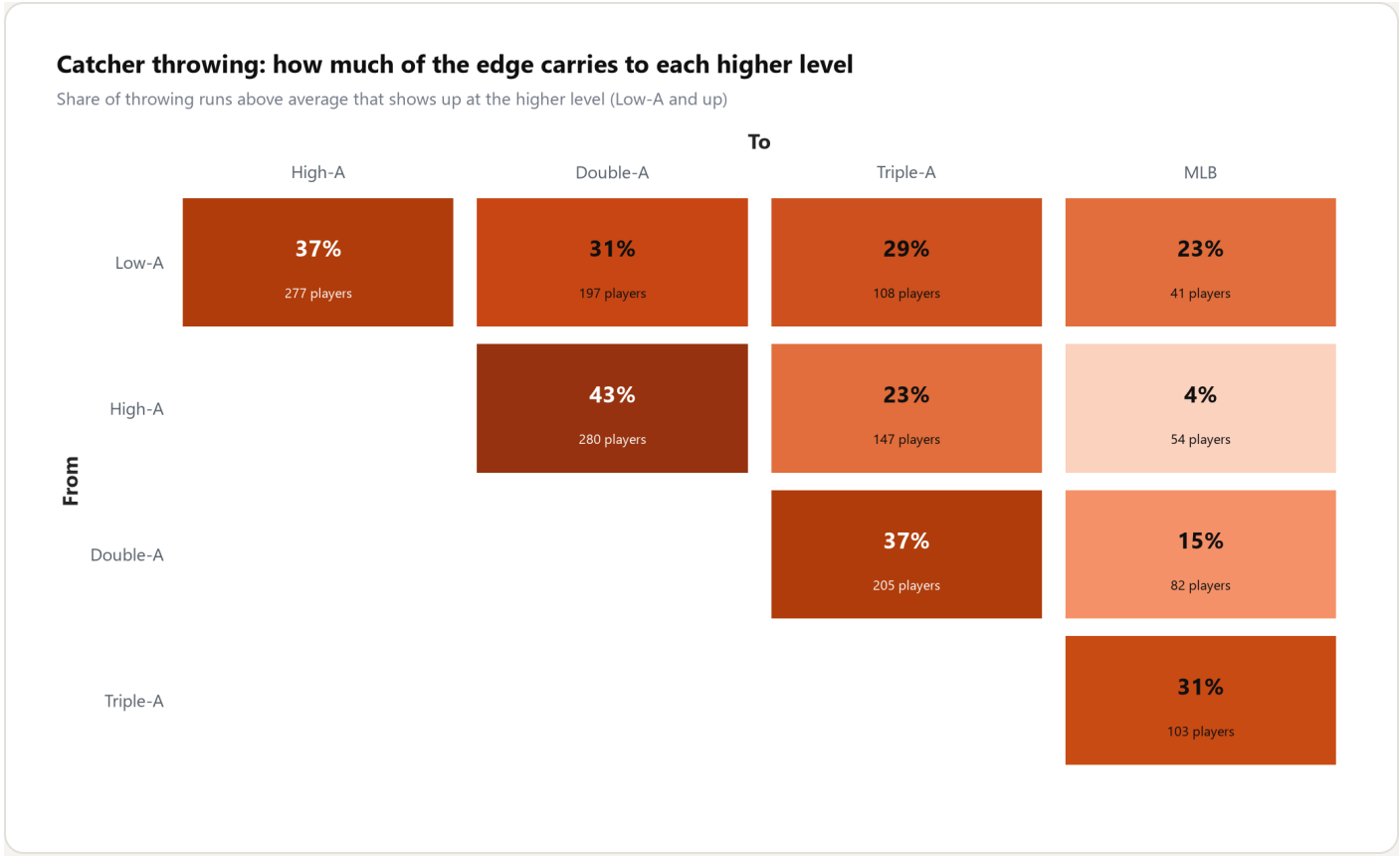
These grids cover any pair of levels, not just the next step up. They use players who played at both levels and compare their full record at each.



Catcher framing: how much of the edge carries to each higher level

Share of framing runs above average that shows up at the higher level

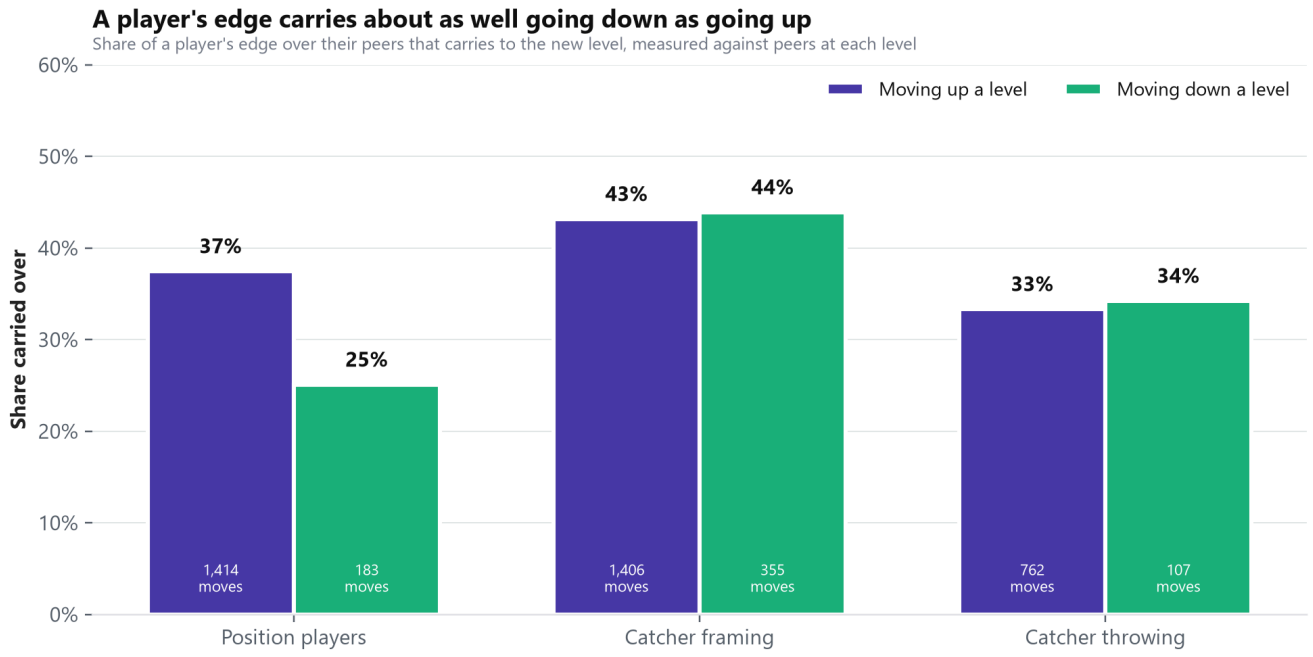




How to read the grids: find the level a player is at now on the left, then the level you're projecting to along the top. A cell reading 30% means a player who is 10 runs per 100 above average should be expected to be about 3 above average at that level. Grey cells don't have enough players who played at both levels to give a reliable answer.

- **Between full-season levels** (Low-A and up), position players keep about 11%–48% of their edge from one level to any higher one. For example, a position player at +3.0 in Low-A projects to about +0.7 in Double-A.
- **For catchers,** a framer at +0.3 in Triple-A projects to about +0.11 in MLB, and a thrower at +5 runs per 100 steal attempts in Double-A projects to about +1.6 in Triple-A.
- **Rookie-ball numbers barely translate.** Only 8 position players in the data played at both ROK DSL and Double-A, too few to say anything. For the 22 catchers who did, DSL framing provided essentially no information about their Double-A framing.
- **By position:** the interactive calculator in the web version of this article can also translate one position at a time. Splitting by position leaves fewer players per cell, so only 43 of the 147 position-and-level combinations have enough players to translate.

Moving down a level



The same check works in reverse, using players who were sent down a level (183 position player moves, 128 of them from MLB to Triple-A, and 355 catcher framing moves). To compare fairly, the chart measures each player against their peers at each level, because the spread of raw numbers is wider at lower levels.

- **Catchers keep about the same share either way:** about 43% of their framing edge going up and 44% going down, and about 33% and 34% for throwing.
- **Position players keep a little less going down** (25% against 37%), on a smaller sample made up mostly of players sent down from MLB.
- **Good defenders stay good after a demotion.** Catchers who were top-20% framers before being sent down were above average at the lower level 87% of the time.

If facing tougher competition were what shrinks a player's edge, players moving down would keep more of theirs. They keep about the same share (position players even a little less). That points to the main reason the edge shrinks after any move: only about a third of one season's number reflects lasting skill, and the rest doesn't repeat, whichever direction the player goes. So a translation should pull a number toward average by about the same amount up or down the ladder.

Switching to a neighbouring position

Switch	Players	Same players' change	Edge that carries over
2B → SS	115	-0.5	39%
SS → 2B	115	+0.5	31%
SS → 3B	52	+1.5	42%
3B → SS	52	-1.5	45%
LF → CF	174	-1.5	37%
CF → LF	174	+1.5	29%
CF → RF	229	+1.7	38%
RF → CF	229	-1.7	37%

This uses players who qualified at two neighbouring positions in the same season and level, so the player and the competition are the same and only the position changes. "Same players' change" is how much better or worse those players did at the new spot, in runs per 100 chances. "Edge that carries over" is how much of a player's advantage over their peers at the old spot shows up at the new one.

- **The harder spot costs runs.** The same players average about 1.6 runs per 100 chances less in CF than in the corners and 1.5 less at SS than at 3B. The step from 2B to SS is smaller, about 0.5.
- **Some of the edge carries over.** A position player above average at one spot keeps part of that edge at the neighbouring one, but less than half.
- **1B ↔ 2B:** only 5 players played both in the same season and level, too few to translate.

What This Means for Player Development

- **Look at rank, not the raw number.** Where a player ranks among their peers tells you more about the next level than their runs-saved total does.
- **Expect big numbers to come down.** A player well above average at one level will usually still be above average at the next, but by less.
- **Be careful with rookie-ball numbers.** They swing the most, and they barely translate to the upper levels.
- **Minor-league framing is a strong signal.** Catchers who frame well in the minors usually keep framing well.
- **Judge a catcher's arm by pop time, in the context of their staff.** A catcher's throwing runs rise and fall with how quickly their pitchers get the ball to the plate.
- **Don't expect promotions to fix defense.** Poor position players keep moving up, so work on defense in the lower levels, while players are still gaining ground.
- **Position changes show up quickly.** Expect a player's grade to drop at a harder position and rise at an easier one.

Data and Methods

Caveats

- **Who's included.** A position player needs at least 100 chances at one position in a season at a level, and a catcher needs at least 500 called pitches. These cutoffs make sure there's enough data to include a player-season. They don't make one season's number reliable on its own.
- **The MLB comparisons only look back.** They include players who reached MLB with a complete minor-league record. They say nothing about players who were released, stalled, or never reached the cutoffs, and they use a single season (2026), not full careers.
- **Not cause and effect.** The age and position results show patterns, not causes. For example, a team may move a player to an easier position because they were already losing range.
- **Framing only for catchers.** The framing results don't account for the pitching staff, umpires, or changes in a catcher's receiving technique.
- **Throwing tracking is thin.** Throwing covers 2016–2026, Low-A to MLB (the source has one combined rookie code, which can't be split into DSL and Complex League like the rest of this report, so rookie ball is left out). Pop time is tracked on only 11% of steal attempts (2021–2026), arm strength on 21% and the pitcher's time to the plate on 14%, so the steal model and grid rest on a few thousand attempts.
- **The Position Players section leaves out catchers.** Catchers never reach the 100-chance fielding cutoff, so that section covers the other seven positions, and catchers get their own section.

Definitions

- **FRV/100** ("runs saved per 100 chances"): SFG-modeled defensive run value (`-runs_plus_minus`) summed over a player-season at a level and divided by opportunities, times 100. Position player opportunities are modeled plays; framing opportunities are called pitches (`ball_called` / `strike_called`).
- **Percentile** ("rank among peers"): the percentile rank of FRV/100 within the same season and level (and, for position players, position).
- **Spread**: the 90th minus the 10th percentile of FRV/100 at a level. For position players it goes from 11.47 in ROK DSL to 5.03 in MLB. For framing it goes from 1.12 to 0.30. The median

qualified player has 118 fielding chances in ROK DSL against 218 in MLB, and a catcher sees 1,142 called pitches against 3,668, which is part of why the lower levels are noisier.

Last Data Update

2026-09-26.

Code and Queries

GitHub repository: isardinasfg/defensive-run-value-translations.